

Speech-Based Detection of Neurological Diseases Using Audio and Transcripts

Introduction

Neurological disorders such as Parkinson's disease, dysarthria, and dementia often affect speech, making speech a useful signal for early, low-cost, and non-invasive screening. Changes in speech can appear before a condition is fully recognized clinically, so a speech-based system could help support earlier detection. Because audio is much easier to collect than many other clinical measurements, this approach could also be more practical for remote use and broader access.

Current work in neurological disease detection often uses brain imaging, neurological exams, and other medical data to achieve strong results. While those approaches can be effective, they are also more expensive, less accessible, and harder to use for large-scale or remote screening. My goal in this project was to focus mainly on speech audio so that detection could be easier to use, more widely available, and better suited as an early warning tool that suggests when further medical evaluation may be needed.

The objective of this project was to build machine learning models that detect Dementia, Dysarthria, and Parkinson's disease from speech recordings and combine them into a real-time detection system for .wav files. I trained separate disease-specific models, compared multiple approaches, and built a final pipeline that uses audio-first modeling for Parkinson's disease and dysarthria, while using a combined transcript-and-audio approach for dementia.

Methods and Materials

Data

This project used multiple publicly available speech datasets that were organized locally into disease-specific folders for Dementia, NoDementia, Dysarthria, Parkinson's disease, and healthy control speech. Additional Parkinson's-related datasets, including MDVR-KCL, UCI_189, and UCI_301, were also included to support both classification and regression tasks. For the dementia portion of the project, manually prepared transcript files were used alongside the audio recordings so that both speech content and acoustic information could be modeled together.

The main files used to run and reproduce the project were `train_dementia.ipynb`, `train_dysarthria.ipynb`, `train_parkinsons.ipynb`, `train_multiclass.ipynb`, `real_time_detection.py`, and `run_real_time_detection_all_data.py`.

Processing and Modeling

All audio recordings were standardized through the same preprocessing pipeline so that differences in recording format or quality would not affect the models more than the actual disease-related speech patterns. Each .wav file was loaded, converted to mono when needed, normalized, resampled to 16 kHz, and trimmed to reduce silence. This made the recordings more consistent, reduced unnecessary volume differences, aligned the scale across all audio files, and improved pause and voicing measurements.

After preprocessing, each audio file was converted into a set of numeric features that described how the speech sounded. These features measured things such as how long the speech lasted, how loud it was, how often the speaker paused, how steady the voice was, and other patterns related to tone, clarity, and sound shape. Together, they captured changes in loudness, stability, timing, articulation, and overall speech structure.

For dysarthria and the main Parkinson's task, these audio features were used with classical machine learning models such as logistic regression, random forest, and extra trees. The notebooks compared these models using accuracy, balanced accuracy, precision, recall, and F1 score, and the strongest model was selected. The Parkinson's notebook also included additional tasks from MDVR-KCL, UCI_301, and UCI_189, allowing both classification and regression experiments.

Dementia required a different approach because the problem was not only how the speech sounded, but also what was being said. Since audio-only results were weaker, the final dementia notebook matched transcripts to the dementia and no-dementia recordings. The transcript text was converted into text embeddings and simple text statistics. These embeddings turned the transcript into numeric patterns that helped the model capture meaning, word usage, and overall structure. These transcript features were then combined with the output probability of the audio dementia model to create a multimodal classifier. Speaker-level train, validation, and test splits were used so that clips from the same speaker would not appear in multiple parts of the dataset. This reduced leakage and made the evaluation more realistic, since the model had to generalize to speakers it had not already seen. In the final real-time pipeline, transcript text was also used for dementia whenever it was available, which helped improve performance beyond audio alone.

Experimental Procedure

The project workflow was designed so that each disease could first be trained separately and then integrated into a final detection system. The datasets were organized into disease-specific folders so that audio files and, for dementia, transcript files could be matched correctly during

training. Each .wav recording was then preprocessed and converted into extracted audio features, while the dementia transcripts were converted into text-based features and combined with the audio results.

After preparation, multiple candidate models were trained for each disease and compared using accuracy, balanced accuracy, precision, recall, and F1 score. The strongest model for each task was selected and saved in the Models/ folder. These saved models were then used in the final real-time detection pipeline to test overall end-to-end system behavior on available audio files.

To replicate the project, the datasets must be placed in the same folder structure, the disease-specific notebooks must be run to retrain the models, and the saved models can then be tested using `real_time_detection.py` or `run_real_time_detection_all_data.py`.

Results

Disease-Specific Classification Results

The best-performing model for each main disease task is shown in Table 1. Parkinson's disease and dysarthria both achieved very high performance using audio-based features alone, while dementia achieved its best result only after combining transcript and audio information. These results show that Parkinson's and dysarthria were highly separable from speech audio alone, while dementia required text-based modeling. This supports the overall conclusion of the project that different disorders needed different modeling strategies.

Table 1: Disease-Specific Classification Results

Task	Best Model	Accuracy	Precision	Recall	F1 Score
Dementia	Text + audio model	0.9394	0.9200	1.0000	0.9583
Dysarthria	Logistic Regression	0.9713	0.9531	0.9559	0.9545
Parkinson's	Logistic Regression	0.9945	0.9496	1.0000	0.9742

Dementia Results

The final dementia model performed best when transcript features were combined with the audio model output. The confusion matrix in Figure 1 shows that all dementia speakers in the held-out test split were detected correctly, while 8 of 10 no-dementia speakers were also classified correctly.

Figure 1: Dementia Confusion Matrix

confusion matrix		
	Pred NoDementia	Pred Dementia
NoDementia	8	2
Dementia	0	23

Dysarthria and Parkinson's Results

Figures 2 and 3 show the confusion matrices for the best dysarthria and Parkinson's models. For dysarthria, the model correctly classified 723 non-dysarthria samples and 325 dysarthria samples, with only 31 total errors. For Parkinson's, the model correctly classified all 132 Parkinson's samples and 1130 non-Parkinson's samples, with only 7 false positives and no false negatives. These results indicate that both disorders produced strong and consistent speech patterns that were well captured by the extracted audio features.

Figure 2: Dysarthria Confusion Matrix

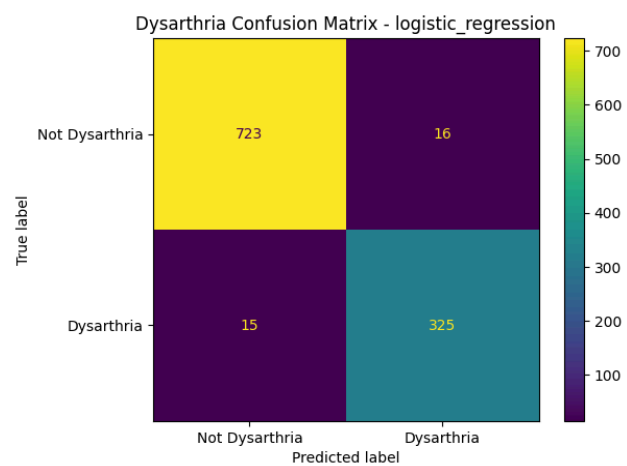
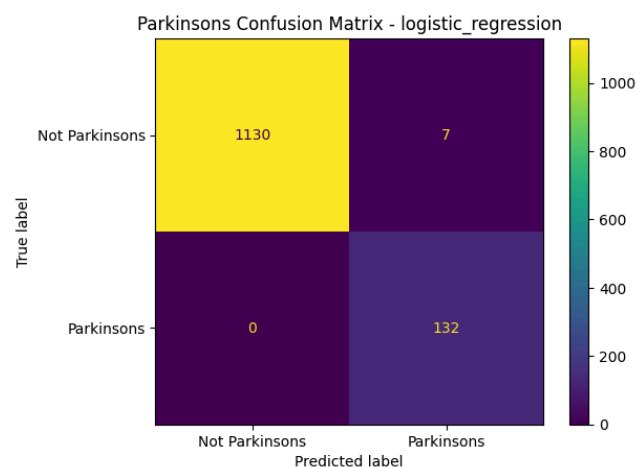


Figure 3: Parkinson's Confusion Matrix



Real-Time Detection Pipeline

The final script `run_real_time_detection_all_data.py` was used as an end-to-end pipeline check over the available project data. It evaluated 5078 total files and produced an overall accuracy of 0.9996. In addition to accuracy, the confidence breakdown showed that most predictions were made with very high confidence: 4704 were labeled Elite, 219 were Excellent, and 67 were Great. Smaller groups fell into Good, Decent, Weak, and Bad, with the lowest-confidence predictions being the most likely to contain errors.

This confidence distribution is important because it shows that the system was not only accurate on the available data, but was also usually highly confident when making correct predictions. Lower-confidence outputs can therefore be interpreted as cases that may need more careful review.

Discussion

The results showed that different disorders needed different modeling strategies. Parkinson's disease and dysarthria worked very well with audio-based features alone, while dementia performed best when transcript information was combined with audio. This showed that dementia depends not only on how speech sounds, but also on what is being said.

The main challenges were inconsistent public data quality, class imbalance, and the difficulty of dementia when treated as an audio-only problem. Dementia required multiple redesigns before the final multimodal approach worked well. Parkinson's and dysarthria were more stable overall, but they still required careful preprocessing and feature selection to handle differences in recording quality. Another challenge was confidence interpretation, since a prediction can be correct even when the confidence is lower, so the final detector separated stronger predictions from weaker ones.

The project also has limitations. The datasets are smaller and more controlled than real-world clinical data, so strong results here may not fully transfer to new speakers, microphones, or recording environments. In addition, the dementia model depends on transcript quality, since the transcripts were generated through an automated script rather than manual clinical transcription. Even with these limitations, the project shows that speech-based machine learning can still be a useful and accessible screening direction. Future work could include testing on external data, improving confidence calibration, and adding a review category for uncertain cases.

Individual Contribution

I completed all parts of this project independently, including data organization, preprocessing, feature extraction, transcript generation and matching, model training, evaluation, real-time detection, and final report preparation.